

In Situ Imaging of a Single-Atom Wave Packet in Continuous Space

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We report on the imaging of the *in situ* spatial distribution of deterministically prepared single-atom wave packets as they expand in a plane, finding excellent agreement with the scaling dynamics predicted by the Schrödinger equation. Our measurement provides a direct and quantitative observation of the textbook free expansion of a one-particle Gaussian wave packet, which we believe has no equivalent in the existing literature. Second, we utilize these expanding wave packets as a benchmark to develop a protocol for the controlled projection of a spatially extended wave function from continuous space onto the sites of a deep optical lattice and subsequent single-atom imaging using quantum gas microscopy techniques. By probing the square modulus of the wave function for various lattice ramp-up times, we show how to obtain a near-perfect projection onto lattice sites. Establishing this protocol represents a crucial prerequisite to the realization of a quantum gas microscope for continuum physics. The method demonstrated here for imaging a wave packet whose initial extent greatly exceeds the pinning lattice spacing, is designed to be applicable to the many-body wave function of interacting systems in continuous space, promising a direct access to their microscopic properties, including spatial correlation functions up to high order and large distances.

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In 1924, Louis de Broglie proposed his theory of electronic waves, where he merged the concepts of particles and waves through the notion of wave-particle duality, which became foundational to the theory of quantum mechanics. A massive particle is associated with a wave function whose dynamics is governed by the Schrödinger equation. This was confirmed in the following years in a number of seminal experiments, such as the celebrated observation of electron diffraction by Davisson and Germer [1], and the diffraction of helium atoms off a crystal surface by Estermann and Stern [2], demonstrating the wave nature of composite particles. Since these historic measurements, wave behavior of massive particles has been evidenced in a wide range of experiments: wave packets were revealed through interference, diffraction, or scattering [3–15], and for particles of increasing size and complexity, from elementary particles such as electrons [16,17] to composite systems such as atoms, clusters, and molecules composed of up to 2000 atoms [18,19].

Another class of experiments has allowed probing the spatial distribution of individual wave packets. For instance, in quantum dots, the static spatial probability density of single electrons has been measured directly or indirectly via scanning tunneling spectroscopy [20,21]. In hybrid atom-ion systems, the vibrational wave packet dynamics of single molecules within their molecular potential was directly observed using high-resolution ion microscopy [22]. On ultracold atom platforms, the tunneling dynamics of single-particle wave packets in a periodic potential were observed with various imaging methods [23–28]. In atom tweezer

harmonic traps, the squared modulus of the wave function in momentum space has been probed for few-particle [29] and single-particle states [30].

Here, we prepare single atoms near the ground state of harmonic oscillator wells and probe the associated Gaussian wave packets *in situ* by following their dynamics upon release from the trap as they expand in a plane. By varying the initial momentum spread of the single-atom wave packets, we observe dynamics that is in quantitative agreement with the prediction from the Schrödinger equation. Our measurement represents a pristine observation of the textbook ballistic expansion of a single-atom Gaussian wave packet in real space. This is realized by performing quantum gas microscopy [31–34] after projecting the wave function freely expanding in space onto a deep optical lattice [35–37]. Using the known single-particle wave packet expansion, we quantitatively determine a protocol for controlled projection of a wave function evolving in continuous space and reliable pinning of the corresponding atom for imaging. With this, we achieve a crucial prerequisite to extend the use of quantum gas microscopy from lattice systems to continuous-space many-body systems, offering direct access to their spatial correlation functions up to high order and large distances.

We start with a dilute ensemble of lithium 6 (⁶Li) atoms prepared near the ground state of the wells of a deep optical lattice. The optical lattice is generated by the self-interference of a laser beam in the horizontal *x*-*y* plane forming three arms crossing at 120° angles [Fig. 1(a)], with a vertical polarization that results in a triangular lattice

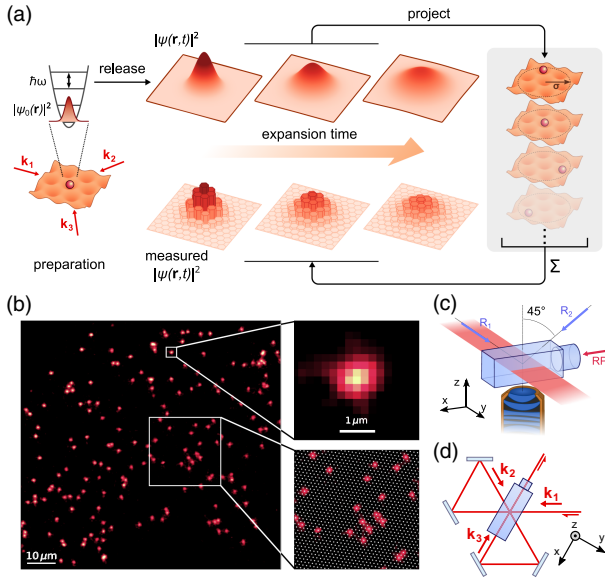


FIG. 1. Preparation and *in situ* imaging of single-atom wave packets. (a) Measurement scheme: Individual atoms are prepared close to the harmonic oscillator ground state of individual sites of a triangular optical lattice created by a self-interfering laser beam with wave vectors $\mathbf{k}_1$, $\mathbf{k}_2$, and $\mathbf{k}_3$. Wave packets initially trapped in the lattice wells, characterized by a Gaussian probability density distribution $|\psi_0(\mathbf{r})|^2$, are released in a plane, allowing them to expand for a given time. For imaging after expansion, the lattice is quickly ramped up again, projecting the wave packet, and Raman sideband cooling is applied to pin the atom on a single site. Resulting atomic positions are recorded through site-resolved fluorescence imaging. From many repetitions of identically prepared wave packets we create histograms of the projected positions with a discretization given by the lattice structure, resulting in a measured probability distribution $|\psi(\mathbf{r}, t)|^2$. (b) Experimental single-atom resolved image. The top-right panel shows a subregion containing an individual atom. The bottom-right panel displays an enlarged region of the image over which the reconstructed triangular lattice structure with a spacing of $a_L = 709$ nm is shown as white dots. (c) Experimental configuration of the oblate optical dipole trap confining the atoms to a two-dimensional plane, the Raman beams (R_1 , R_2 , and RP) used for cooling and imaging, and the microscope objective. (d) Top view of the experimental configuration, showing the geometry of the optical lattice beam.

geometry [38]. Close to the bottom of the wells, the potential seen by the atoms is well approximated [39] by a harmonic oscillator $U(x, y) \simeq \frac{1}{2}m\omega^2(x^2 + y^2)$, with $\omega \simeq 2\pi \times 1$ MHz, leading to eigenstates $|n_x, n_y\rangle$ and associated energy $(n_x + n_y + 1)\hbar\omega$. In the vertical direction (z axis), the atoms are confined in a single plane using a laser-light sheet [40]. We typically load a few tens of atoms over the ~ 6000 available lattice sites in order to sparsely populate the lattice. The interparticle spacing is on the order of $10\ \mu\text{m}$, more than 10 times the lattice spacing $a_L = 709$ nm, such that each occupied well contains no more than one atom. The loading of tens of independent

atoms allows us to obtain as many realizations of the single-particle expansion dynamics in a single experimental run.

In order to bring the atoms near the ground state of the lattice wells, we apply Raman sideband cooling (RSC), as described in the End Matter. The photons that are spontaneously emitted during this laser cooling process serve as a fluorescence signal to detect each atom while it is held in a given site. This dual-purpose approach of RSC is routinely used in quantum gas microscopy [33,34], where it was developed for the study of fermionic systems in optical lattices [45–47]. Our imaging method is identical to quantum gas microscopy, except that we apply it to probe wave packets in continuous space. In Fig. 1(b), we show a typical single-atom image obtained with our apparatus.

In this first step, we therefore not only prepare an ensemble of single-atom Gaussian wave functions near the ground state of harmonic oscillators, but also have an image of their initial positions with a resolution at the level of a single lattice site. Following this initialization phase, we turn off all RSC laser beams, and adiabatically ramp down the lattice depth to a variable fraction U_0 of its initial value U_{max} , which allows us to control the frequency of the harmonic oscillator $\omega \propto \sqrt{U_0}$ while maintaining the wave packets in their initial vibrational level. We then suddenly turn off the lattice beams, keeping the light sheet potential on, to let the atomic wave packets expand in the x - y plane for a variable time t . We subsequently turn the lattice back on to its maximal depth using an optimal ramp time (see below), and perform RSC again to pin the atoms. This step projects the free atoms to the nearest lattice site after expansion. We thus obtain a second image displaying the final positions of the atoms. Assuming no bias is introduced upon pinning from continuous space and imaging, other than a discretization of space, each atom on the second image will be detected at a certain position, sampling the probability density—i.e., the modulus squared—of the wave function after release. By relating each atom of the second image to its original lattice site in the first image from many realizations (see End Matter), we obtain histograms in position space representing the probability density of the expanding wave packets at variable time t .

We repeat the measurement above for initial wave packets in traps of different frequencies ranging from $\omega = 2\pi \times 260(15)$ to $2\pi \times 760(30)$ kHz, by varying U_0 from 7% to 61% of the maximum value. In a given run we typically produce 20 to 50 independent but identically prepared wave packets. For each preparation and expansion time t we record 100 image pairs (see End Matter) from which we obtain an averaged histogram reflecting the probability density. In Fig. 2, we show experimentally measured probability densities for each of these wave packets at different times of their dynamics, alongside the extracted evolution of their Gaussian width $\sigma(t)$, which shows the expected linear behaviour in t at long times [see Fig. 2(c)]. Indeed, for a wave packet prepared with an average vibrational number $\langle n \rangle = (\langle n_x \rangle + \langle n_y \rangle)/2$, the

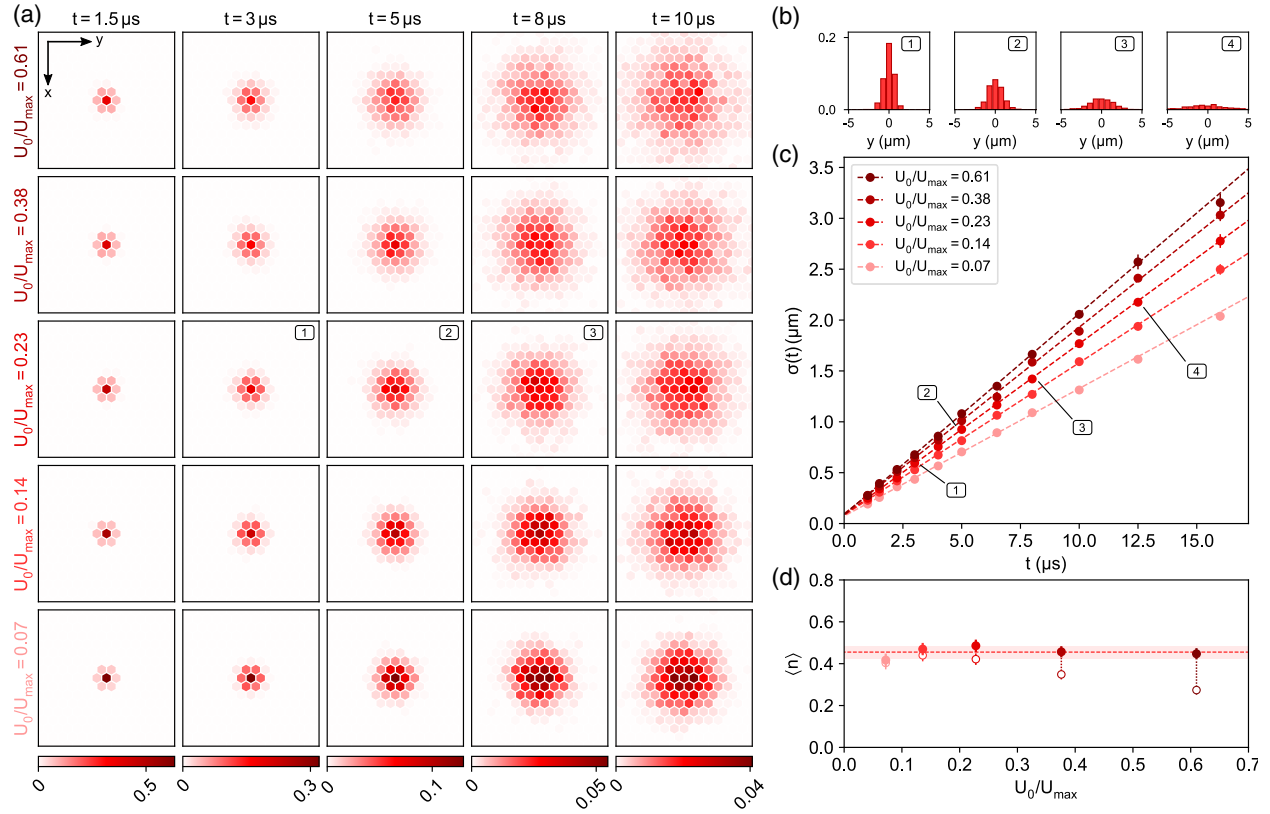


FIG. 2. Expansion dynamics of a single-atom Gaussian wave packet. (a) Probability density distribution as a function of expansion time t for wave packets prepared at different lattice depths U_0 before release. Hexagons represent the lattice sites of the triangular pinning lattice, with the central site indicating the position of each atom in the first single-atom image. The histograms are obtained after 100 measurements, each with around 20 to 50 identically prepared single-atom wave packets. (b) Cuts of the two-dimensional histograms at $U_0 = 0.23 U_{\max}$ for $t = 3, 5, 8$, and $12.5 \mu\text{s}$. (c) Data points show the width of the Gaussian wave packet (σ) extracted from the probability density distribution as a function of t for each preparation lattice depth. Error bars indicate the standard error of the maximum likelihood estimate and are typically smaller than the symbol size. Dashed lines represent linear fits to the experimental data. (d) Solid points show the average harmonic oscillator eigenvalues $\langle n \rangle = [0.42(3), 0.47(3), 0.49(3), 0.46(3), 0.45(3)]$ for each preparation, corresponding to the lattice depth $U_0/U_{\max} = 0.072(5), 0.136(5), 0.228(7), 0.376(10)$, and $0.61(2)$. The open points are the values extracted from the linear fits in (b) using Eq. (2) before the correction due the noninstantaneous release is taken into account (see text). The horizontal red dashed line together with the light red band indicate the average of all five values $\langle n \rangle = 0.46(3)$.

Schrödinger equation predicts that its width $\sigma(t)$ evolves as

$$\sigma^2(t) = \sigma^2(0) + \frac{\Delta p^2(0)}{m^2} t^2 \quad (1)$$

$$= (2\langle n \rangle + 1) \frac{\hbar}{2m\omega} [1 + \omega^2 t^2]. \quad (2)$$

In the long time limit $\omega^2 t^2 \gg 1$, this simplifies to $\sigma(t) = \sqrt{\frac{\hbar\omega}{2m}} \sqrt{2\langle n \rangle + 1} \cdot t$. The slope of $\sigma(t)$ in this regime provides a direct measurement of $\langle n \rangle$. In Fig. 2(d), we report the measured values of $\langle n \rangle$ for all trap depths U_0 , which all agree to better than 8% with the overall average value of $\langle n \rangle = 0.46(3)$. The consistency among the values of $\langle n \rangle$, combined with the linear expansion, confirms the faithful measurement of the dynamics predicted by the Schrödinger equation. For a thermal distribution of the

vibrational state populations, this corresponds to $\sim 50\%$ of atoms in the ground state $|n_x = 0, n_y = 0\rangle$.

These values include a small correction due to the non-instantaneous release of the wave packet. Indeed, the turn-off of the lattice power follows a decaying sigmoid with a characteristic time of ~ 250 ns. For the turn-off to be considered instantaneous, this timescale should be negligible compared to the harmonic oscillator period $T = 2\pi/\omega$. Otherwise, the trapped wave packet first experiences a continuous opening of the harmonic trap before release, resulting in an effective reduction of the kinetic energy, and a lower apparent $\langle n \rangle$ during the free expansion. We quantitatively account for this effect and predict downshifts of the average motional number $\delta n = [0.014(1), 0.030(1), 0.064(2), 0.108(3), 0.173(4)]$ for increasing values of U_0 [40], with the largest impact for the two highest values of U_0 , corresponding to $T = 1.65(10)$ and $T = 1.3(1) \mu\text{s}$. Correcting for this shift

results in the actual values of $\langle n \rangle$. Our ability to quantitatively distinguish this subtle effect demonstrates the high level of control of our setup and the precision of our imaging protocol.

The agreement between the measured and predicted wave packets at each instant of the dynamics serves as a validation of our procedure for projecting a spatially extended wave function from continuous space onto the sites of a deep optical lattice, which has never been addressed quantitatively. In most quantum gas microscope experiments, atoms are initially loaded in the lowest band of a shallow optical lattice, and the pinning is performed by ramping up the lattice power according to the adiabaticity criterion $\dot{\omega}/\omega^2 \ll 1$. By contrast, pinning atoms from the continuum leads to a more complex projection dynamics of the wave function [48]. Although it is impossible to be strictly adiabatic—due to the absence of an initial energy gap—one can still mitigate the impact of diabatic processes. This requires setting a lower limit on the lattice ramp-on time τ , as an instantaneous ramp-on would lead to a non-negligible projection onto higher bands, where atoms are less efficiently cooled. The final lattice frequency ω sets a suitable timescale for this requirement, i.e. $\omega\tau \gg 1$. On the other hand, a reliable measurement of the initial atom positions requires the lattice to be ramped fast enough to prevent any significant expansion of the wave packet over the ramp-up time τ . Ideally, atoms should be projected to the nearest site upon pinning, which sets the condition that $v\tau \lesssim a_L$, where v is a characteristic velocity of the system. The lattice ramp time is therefore constrained by the double inequality $\omega^{-1} \ll \tau \lesssim a_L/v$.

To quantitatively address this question, we performed the same type of wave packet expansion measurement, varying the ramp up time of the lattice τ , for a fixed expansion time t . As figures of merit for the reliability of the pinning, we extract the Gaussian width σ , and the probability p_{ideal} to be pinned to the closest lattice site for different ramp up times. The probability p_{ideal} is directly provided by our maximum likelihood analysis [40]. In Fig. 3, we present a typical optimization measurement showing that p_{ideal} first increases rapidly to reach a maximum value of 99% for $\tau = 2\text{--}5\ \mu\text{s}$ and starts to slowly decrease for longer times. On the other hand, the Gaussian width σ is essentially constant up to $\tau = 5\ \mu\text{s}$ and then follows a linear increase. In terms of the previously discussed time constraints, σ is expected to grow with τ and hence serves to identify the upper time limit, while keeping $p_{\text{ideal}} \approx 1$ sets the lower time limit. We are thus in a favorable situation where we have a range of values for τ , where p_{ideal} is maximal while σ remains close to its $\tau = 0$ value. This is quantitatively consistent with the double inequality above, which in this specific case yields $0.16\ \mu\text{s} \ll \tau \lesssim 4\ \mu\text{s}$. For all the data presented in the previous section, we have used a fixed ramp time $\tau = 4\ \mu\text{s}$ for the sake of consistency, with a negligible cost on the precision of σ for the highest trap depths.

This study lays the groundwork for quantum gas microscopy in the continuum, by establishing a general requirement on the pinning timescale for the projection of an extended

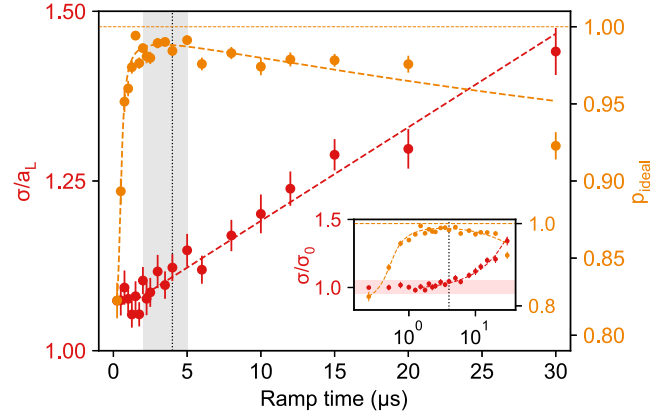


FIG. 3. Continuum pinning dynamics. Normalized Gaussian width σ/a_L (red data points) and probability p_{ideal} for an atom to be pinned to the closest lattice site (yellow data points) as a function of the lattice ramp time, for $U_0/U_{\text{max}} = 0.23$ and an expansion time of $4\ \mu\text{s}$. The vertical gray band displays the ramp time interval where p_{ideal} is maximal and the vertical dashed line shows the ramp time used for the data presented in Fig. 2. The yellow dotted line indicates $p_{\text{ideal}} = 1$, representing perfect pinning from continuum. The loss probability remains constant across ramp times ($\approx 0.5\%$). The inset shows a semilogarithmic plot of p_{ideal} , and σ normalized to the initial width, with the horizontal red band indicating a $\pm 5\%$ interval. Dashed lines represent guides to the eye for the respective datasets. Error bars give the standard error of the maximum likelihood estimates.

wave function from continuous space, which is constrained by two limits. While the lower time limit will remain fixed for most experiments, the upper limit depends on the typical particle velocity. For instance, for a 2D degenerate Fermi gas with an inter-particle spacing of a few lattice sites, the typical Fermi velocities are $\sim 0.02\ \text{m/s}$, an order of magnitude smaller than the particle velocities probed here. The upper ramp-up time limit could therefore be relaxed even further in that case, which implies that our projection protocol is directly applicable to such a system. Note that for a given atomic density, lattice frequency and spacing, the above criterion is more favorable for heavier species.

In this work, we observed the textbook expansion dynamics of a one-atom Gaussian wave packet using a new protocol for single-atom-resolved imaging in continuous space. The excellent agreement obtained here with the scaling dynamics predicted by the Schrödinger equation constitutes a crucial benchmark for the reliability of our imaging method. Our work represents a milestone towards applying quantum gas microscopy to continuous-space many-body systems in the near future.

Complementing recent advances in single-atom imaging of tweezer-trapped few fermion systems [29,49,50], our approach opens radically new possibilities for exploring correlated bulk quantum gases with a large number of particles at the microscopic level, for instance, by allowing direct access to spatially resolved correlation functions. In combination with currently available homogeneous potentials [51–54], our imaging technique will facilitate the

search for exotic phases of matter that have been elusive so far, such as the Fulde-Ferrell-Larkin-Ovchinnikov superfluid [55,56] or atomic quantum Hall fluids with fractional statistics [57–59].

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- [1] C. J. Davisson and L. H. Germer, *Proc. Natl. Acad. Sci. U.S.A.* **14**, 317 (1928).
- [2] I. Estermann and O. Stern, *Z. Phys.* **61**, 95 (1930).
- [3] R. F. Egerton, *Physical Principles of Electron Microscopy* (Springer US, Boston, MA, 2005).
- [4] M. N. R. Ashfold, N. H. Nahler, A. J. Orr-Ewing, O. P. J. Vieuxmaire, R. L. Toomes, T. N. Kitsopoulos, I. A. Garcia, D. A. Chestakov, S.-M. Wu, and D. H. Parker, *Phys. Chem. Chem. Phys.* **8**, 26 (2006).
- [5] A. D. Cronin, J. Schmiedmayer, and D. E. Pritchard, *Rev. Mod. Phys.* **81**, 1051 (2009).
- [6] A. S. Stodolna, A. Rouzée, F. Lépine, S. Cohen, F. Robicheaux, A. Gijsbertsen, J. H. Jungmann, C. Bordas, and M. J. J. Vrakking, *Phys. Rev. Lett.* **110**, 213001 (2013).
- [7] H. Rauch and S. A. Werner, *Neutron Interferometry: Lessons in Experimental Quantum Mechanics, Wave-Particle Duality, and Entanglement* (Oxford University Press, New York, 2015).
- [8] M. McDonald, B. H. McGuyer, F. Apfelbeck, C.-H. Lee, I. Majewska, R. Moszynski, and T. Zelevinsky, *Nature (London)* **535**, 122 (2016).
- [9] W. E. Perreault, N. Mukherjee, and R. N. Zare, *Science* **358**, 356 (2017).
- [10] M. Waitz *et al.*, *Nat. Commun.* **8**, 2266 (2017).
- [11] Q. Guan, V. Klinkhamer, R. Klemm, J. H. Becher, A. Bergschneider, P. M. Preiss, S. Jochim, and D. Blume, *Phys. Rev. Lett.* **122**, 083401 (2019).
- [12] E. T. Karamatskos, S. Raabe, T. Mullins, A. Trabattini, P. Stammer, G. Goldsztejn, R. R. Johansen, K. Długołęcki, H. Stapelfeldt, M. J. J. Vrakking, S. Trippel, A. Rouzée, and J. Küpper, *Nat. Commun.* **10**, 3364 (2019).
- [13] B. R. Heazlewood and T. P. Softley, *Nat. Rev. Chem.* **5**, 125 (2021).
- [14] P. Paliwal, N. Deb, D. M. Reich, A. van der Avoird, C. P. Koch, and E. Narevicius, *Nat. Chem.* **13**, 94 (2021).
- [15] T. de Jongh, Q. Shuai, G. L. Abma, S. Kuipers, M. Besemer, A. van der Avoird, G. C. Groenenboom, and S. Y. T. van de Meerakker, *Nat. Chem.* **14**, 538 (2022).
- [16] P. W. Hawkes and E. Kasper, *Principles of Electron Optics: Wave Optics* (Academic Press, London, 1996).
- [17] A. Tonomura, J. Endo, T. Matsuda, T. Kawasaki, and H. Ezawa, *Am. J. Phys.* **57**, 117 (1989).
- [18] Y. Y. Fein, P. Geyer, P. Zwick, F. Kiałka, S. Pedalino, M. Mayor, S. Gerlich, and M. Arndt, *Nat. Phys.* **15**, 1242 (2019).
- [19] K. Hornberger, S. Gerlich, P. Haslinger, S. Nimmrichter, and M. Arndt, *Rev. Mod. Phys.* **84**, 157 (2012).
- [20] E. E. Vdovin, A. Levin, A. Patané, L. Eaves, P. C. Main, Yu. N. Khanin, Yu. V. Dubrovskii, M. Henini, and G. Hill, *Science* **290**, 122 (2000).
- [21] T. Maltezopoulos, A. Bolz, C. Meyer, C. Heyn, W. Hansen, M. Morgenstern, and R. Wiesendanger, *Phys. Rev. Lett.* **91**, 196804 (2003).
- [22] Y.-Q. Zou, M. Berngruber, V. S. V. Anasuri, N. Zuber, F. Meinert, R. Löw, and T. Pfau, *Phys. Rev. Lett.* **130**, 023002 (2023).
- [23] M. Karski, L. Förster, J.-M. Choi, A. Steffen, W. Alt, D. Meschede, and A. Widera, *Science* **325**, 174 (2009).
- [24] C. Weitenberg, M. Endres, J. F. Sherson, M. Cheneau, P. Schauß, T. Fukuhara, I. Bloch, and S. Kuhr, *Nature (London)* **471**, 319 (2011).
- [25] T. Fukuhara, A. Kantian, M. Endres, M. Cheneau, P. Schauß, S. Hild, D. Bellem, U. Schollwöck, T. Giamarchi, C. Gross, I. Bloch, and S. Kuhr, *Nat. Phys.* **9**, 235 (2013).
- [26] M. Genske, W. Alt, A. Steffen, A. H. Werner, R. F. Werner, D. Meschede, and A. Alberti, *Phys. Rev. Lett.* **110**, 190601 (2013).
- [27] P. M. Preiss, R. Ma, M. E. Tai, A. Lukin, M. Rispoli, P. Zupancic, Y. Lahini, R. Islam, and M. Greiner, *Science* **347**, 1229 (2015).
- [28] A. W. Young, W. J. Eckner, N. Schine, A. M. Childs, and A. M. Kaufman, *Science* **377**, 885 (2022).
- [29] M. Holtz, L. Bayha, K. Subramanian, C. Heintze, P. M. Preiss, and S. Jochim, *Phys. Rev. Lett.* **126**, 020401 (2021).
- [30] M. O. Brown, S. R. Muleady, W. J. Dworschack, R. J. Lewis-Swan, A. M. Rey, O. Romero-Isart, and C. A. Regal, *Nat. Phys.* **19**, 569 (2023).
- [31] W. S. Bakr, J. I. Gillen, A. Peng, S. Fölling, and M. Greiner, *Nature (London)* **462**, 74 (2009).
- [32] J. F. Sherson, C. Weitenberg, M. Endres, M. Cheneau, I. Bloch, and S. Kuhr, *Nature (London)* **467**, 68 (2010).
- [33] C. Gross and I. Bloch, *Science* **357**, 995 (2017).
- [34] C. Gross and W. S. Bakr, *Nat. Phys.* **17**, 1316 (2021).
- [35] W. S. Bakr, A. Peng, M. E. Tai, R. Ma, J. Simon, J. I. Gillen, S. Fölling, L. Pollet, and M. Greiner, *Science* **329**, 547 (2010).
- [36] P. T. Brown, E. Guardado-Sanchez, B. M. Spar, E. W. Huang, T. P. Devereaux, and W. S. Bakr, *Nat. Phys.* **16**, 26 (2020).
- [37] S. Buob, J. Höschele, V. Makhalov, A. Rubio-Abadal, and L. Tarruell, *PRX Quantum* **5**, 020316 (2024).
- [38] S. Jin, K. Dai, J. Verstraten, M. Dixmieras, R. Alhyder, C. Salomon, B. Peaudecerf, T. de Jongh, and T. Yefsah, *Phys. Rev. Res.* **6**, 013158 (2024).
- [39] For the sake of clarity we here ignore a slight difference between the frequencies in the x and y directions. This is, however, taken into account in the data analysis.
- [40] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/PhysRevLett.134.083403> for additional details on the experimental setup and the data analysis. The Supplemental Material includes Refs. [41–44].
- [41] D. J. Heinzen and D. J. Wineland, *Phys. Rev. A* **42**, 2977 (1990).

- [42] A. J. Kerman, V. Vuletic, C. Chin, and S. Chu, *Phys. Rev. Lett.* **84**, 439 (2000).
- [43] R. E. Burkard and E. Cela, in *Handbook of Combinatorial Optimization: Supplement Volume A* (Springer, New York, 1999), pp. 75–149.
- [44] J. H. R. Lewis and W. B. Riesenfeld, *J. Math. Phys. (N.Y.)* **10**, 1458 (1969).
- [45] L. W. Cheuk, M. A. Nichols, M. Okan, T. Gersdorf, V. V. Ramasesh, W. S. Bakr, T. Lompe, and M. W. Zwierlein, *Phys. Rev. Lett.* **114**, 193001 (2015).
- [46] M. F. Parsons, F. Huber, A. Mazurenko, C. S. Chiu, W. Setiawan, K. Wooley-Brown, S. Blatt, and M. Greiner, *Phys. Rev. Lett.* **114**, 213002 (2015).
- [47] A. Omran, M. Boll, T. A. Hilker, K. Kleinlein, G. Salomon, I. Bloch, and C. Gross, *Phys. Rev. Lett.* **115**, 263001 (2015).
- [48] J. H. Denschlag, J. E. Simsarian, H. Häffner, C. McKenzie, A. Browaeys, D. Cho, K. Helmerson, S. L. Rolston, and W. D. Phillips, *J. Phys. B* **35**, 3095 (2002).
- [49] M. Holten, L. Bayha, K. Subramanian, S. Brandstetter, C. Heintze, P. Lunt, P. M. Preiss, and S. Jochim, *Nature (London)* **606**, 287 (2022).
- [50] S. Brandstetter, P. Lunt, C. Heintze, G. Giacalone, L. H. Heyen, M. Galka, K. Subramanian, M. Holten, P. M. Preiss, S. Floerchinger, and S. Jochim, *Nat. Phys.* **21**, 52 (2025).
- [51] A. L. Gaunt, T. F. Schmidutz, I. Gotlibovych, R. P. Smith, and Z. Hadzibabic, *Phys. Rev. Lett.* **110**, 200406 (2013).
- [52] L. Chomaz, L. Corman, T. Bienaimé, R. Desbuquois, C. Weitenberg, S. Nascimbène, J. Beugnon, and J. Dalibard, *Nat. Commun.* **6**, 6162 (2015).
- [53] B. Mukherjee, Z. Yan, P. B. Patel, Z. Hadzibabic, T. Yefsah, J. Struck, and M. W. Zwierlein, *Phys. Rev. Lett.* **118**, 123401 (2017).
- [54] N. Navon, R. P. Smith, and Z. Hadzibabic, *Nat. Phys.* **17**, 1334 (2021).
- [55] P. Fulde and R. A. Ferrell, *Phys. Rev.* **135**, A550 (1964).
- [56] A. I. Larkin and Y. N. Ovchinnikov, *Sov. Phys.-JETP* **20**, 762 (1965).
- [57] N. Cooper, *Adv. Phys.* **57**, 539 (2008).
- [58] T.-L. Ho, [arXiv:1608.00074](https://arxiv.org/abs/1608.00074).
- [59] C. Repellin, T. Yefsah, and A. Sterdyniak, *Phys. Rev. B* **96**, 161111(R) (2017).

End Matter

Appendix—The diagrams of the experimental sequence and RSC are shown in Figs. 4(a) and 4(b). The RSC scheme operates as follows: a two-photon process drives

a hyperfine transition $|F = 3/2, n_i\rangle \rightarrow |F = 1/2, n_i - 1\rangle$ that reduces the vibrational level n_i (with $i = x, y$) of atoms in each lattice site, while a third near-resonant

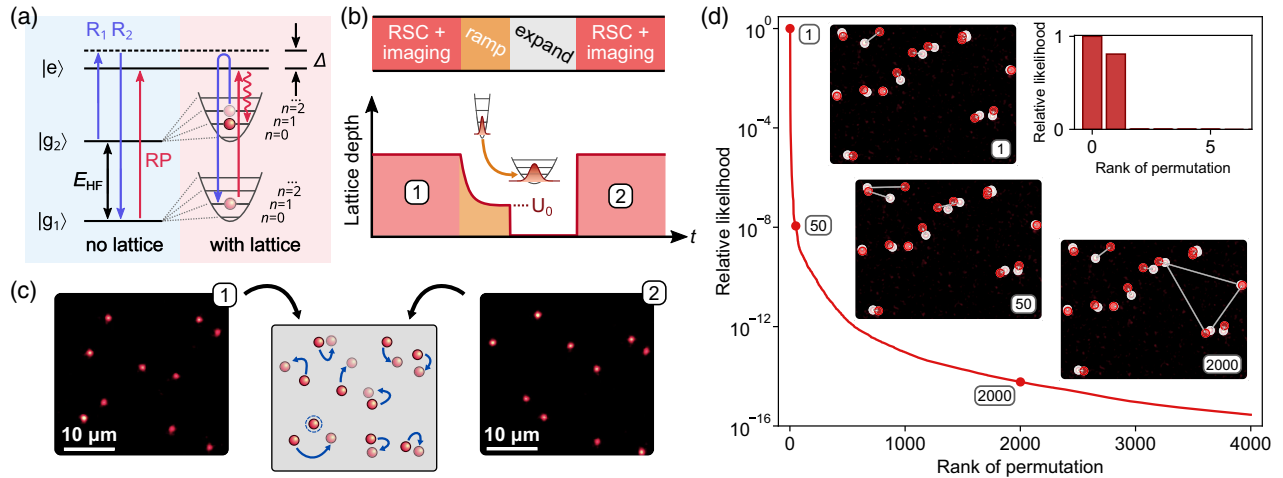


FIG. 4. Experimental sequence and image analysis. (a) Raman sideband cooling scheme for ^6Li atoms. The two Raman beams (R_1 and R_2) are blue detuned by $\Delta = h \times 3$ GHz with respect to the first electronically excited state $|e\rangle$ and connect the two hyperfine levels $|g_1\rangle$ and $|g_2\rangle$ in the ground state manifold, split by $E_{\text{HF}} = h \times 228.2$ MHz. A resonant repumper (RP) beam connects $|g_1\rangle$ and $|e\rangle$. In the presence of a deep optical lattice, RSC brings the atoms to lower harmonic oscillator eigenstates, indicated by eigenvalue n , while producing fluorescence photons for single-particle imaging through the microscope objective, serving as both a preparation and detection method. (b) Experimental sequence for the preparation, expansion, and pinning of the single-atom wave packets. After the wave packets have been localized in a deep optical lattice and their initial positions recorded in a first image (1), RSC is turned off and the lattice depth is adiabatically ramped down to a variable value U_0 to adjust the width of their initial momentum distribution. The lattice is then suddenly switched off and the wave packets expand for a time t after which we take a second image (2) to record the new atom positions. (c) Two single atom images taken in a single experimental realization (left and right panels). The center panel schematically shows the most likely assignment of how the atoms moved between the two images. (d) Relative likelihood of the 4000 most likely assignments based on a combined likelihood computation, for $\omega = 2\pi \times 600(30)$ kHz ($U_0 = 0.38U_{\text{max}}$) and an expansion time of 10 μs . The image insets show the assignments ranked 1st, 50th and 2000th. Initial (white dots) and assigned final (red dots) positions are connected by arrows. The top-right inset shows the relative likelihood in a linear scale.

beam excites the atoms in $|F = 1/2, n_i - 1\rangle$ to the upper state $|F' = 1/2\rangle$, from which they in turn decay to the original hyperfine state $|F = 3/2, n_i - 1\rangle$. At the end of one RSC cycle, the motional energy of a given atom has been reduced by one quantum. After a number of cycles, and in the absence of heating sources, this would result in preparing the atom in the ground state wave function of the harmonic oscillator well. The RSC beam parameters have been optimized to reach the highest fidelity while bringing the atoms as close as possible to the vibrational ground state of the lattice wells. Based on the measured RSC beam and lattice parameters, we expect average vibrational numbers $\langle n_x \rangle \approx \langle n_y \rangle \simeq 0.4$ at the end of the cooling procedure [40].

In order to reliably assign each atom detected after expansion to its initial position, we use a self-consistent analysis based on a maximum likelihood estimate, which we describe in detail in the Supplemental Material [40]. Finding the permutation of atoms that maximizes the total likelihood reduces to the linear assignment problem. In Fig. 4(d), we show an example of relative likelihood as a function of permutation rank, showing a rapid decrease of the probability. The experimental parameters were chosen such that the typical distance traveled by each atom is small compared to the interparticle distance, which strongly facilitates the analysis as it ensures that the likelihood decreases quickly as a function of permutation rank.